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Optical device with elements having outwardly-curved opposing leading edges spaced according to a duty cycle, pitch, and critical dimension

Abstract

Embodiments described herein provide for methods of forming optical device structures having continuously increasing or decreasing duty cycles. One embodiment includes a device. The device includes a plurality of optical device structures. Each optical device structure of the plurality of optical device structures includes a plurality of discrete zones. Each discrete zone of the plurality of discrete zones has a critical dimension. The device further includes a plurality of transition zones disposed between two discrete zones of the plurality of discrete zones. The critical dimension of each transition zone of the plurality of transition zones continuously increases or decreases across a length of each optical device structure of the plurality of optical device structures.

Inventors:	Jiang; Jing (Santa Clara, CA), Fu; Jinxin (Fremont, CA)
Applicant:	Applied Materials, Inc. (Santa Clara, CA)
Family ID:	1000008820694
Assignee:	Applied Materials, Inc. (Santa Clara, CA)
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Primary Examiner: Jordan; Andrew

Attorney, Agent or Firm: Patterson + Sheridan, LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority from U.S. Application Ser. No. 63/136,956, filed Jan. 13, 2021, which is hereby incorporated by reference in its entirety.

BACKGROUND

Field

(1) Embodiments of the present disclosure generally relate to optical devices for augmented, virtual, and mixed reality. More specifically, embodiments described herein provide for methods of forming optical device structures having continuously increasing or decreasing duty cycles.

Description of the Related Art

(2) Virtual reality is generally considered to be a computer generated simulated environment in which a user has an apparent physical presence. A virtual reality experience can be generated in 3D and viewed with a head-mounted display (HMD), such as glasses or other wearable display devices that have near-eye display panels as lenses to display a virtual reality environment that replaces an actual environment.

(3) Augmented reality, however, enables an experience in which a user can still see through the display lenses of the glasses or other HMD device to view the surrounding environment, yet also see images of virtual objects that are generated for display and appear as part of the environment. Augmented reality can include any type of input, such as audio and haptic inputs, as well as virtual images, graphics, and video that enhance or augment the environment that the user experiences. As an emerging technology, there are many challenges and design constraints with augmented reality.

(4) One such challenge is displaying a virtual image overlaid on an ambient environment. Optical devices including waveguide combiners, such as augmented reality waveguide combiners, and flat optical devices, such as metasurfaces, are used to assist in overlaying images. Generated light is propagated through an optical device until the light exits the optical device and is overlaid on the ambient environment. Optical device structures of the optical devices include multiple discrete zones. However, light is lost by the sudden transition between two discrete zones and therefore decreases the optical performance of the optical devices.

(5) Accordingly, what is needed in the art are improved methods of forming optical device structures having continuously increasing or decreasing duty cycles.

SUMMARY

(6) In one embodiment, a device is provided. The device includes a plurality of optical device structures. The plurality of optical device structures have a critical dimension across a length of each optical device structure of the plurality of optical device structures. Each optical device structure of the plurality of optical device structures includes a plurality of discrete zones. The device further includes a plurality of transition zones disposed between two discrete zones of the plurality of discrete zones. The critical dimension of each transition zone of the plurality of transition zones continuously increases or decreases across the length of each optical device structure of the plurality of optical device structures.

(7) In another embodiment, a device is provided. The device includes a plurality of optical device structures. The plurality of optical device structures have a critical dimension across a length of each optical device structure of the plurality of optical device structures. Each optical device structure of the plurality of optical device structures includes a plurality of discrete zones. The device further includes a plurality of transition zones disposed between two discrete zones of the plurality of discrete zones. The critical dimension of each transition zone of the plurality of transition zones continuously increases or decreases across the length of each optical device structure of the plurality of optical device structures. Adjacent optical device structures of the plurality of optical device structures include a pitch. The device further includes a duty cycle defined as the critical dimension of the plurality of optical device structures divided by the pitch. The duty cycle continuously increases or decreases across the length of each optical device structure of the plurality of optical device structures.

(8) In yet another embodiment, a method is provided. The method includes editing a design file. The design file corresponds to a mask. The mask includes a plurality of openings. The plurality of openings have a width that continuously increases or decreases across a length of the plurality of openings. The method further includes generating the mask. The mask is to be disposed over a photoresist disposed over a substrate. The method further includes passing light through the plurality of openings of the mask positioned over the photoresist. The method further includes developing the photoresist to form a patterned photoresist. The patterned photoresist corresponds to the plurality of openings of the mask. The method further includes patterning the substrate or a device material to form a plurality of optical device structures. The optical device structures correspond to the plurality of openings of the mask. The plurality of optical device structures have a critical dimension that continuously increases or decreases across a length of the plurality of optical device structures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only exemplary embodiments and are therefore not to be considered limiting of its scope, and may admit to other equally effective embodiments.

(2) FIG. 1 is a perspective, frontal view of an optical device according to embodiments described herein.

(3) FIGS. 2A and 2B are schematic, top-views of a plurality of optical device structures according to embodiments described herein.

(4) FIGS. 3A and 3B are schematic, cross-sectional views of a substrate according to embodiments described herein.

(5) FIG. 3C is a schematic, top-view of a mask according to embodiments described herein.

(6) FIG. 4 is a flow diagram of a method for forming a plurality of optical device structures according to embodiments described herein.

(7) To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements and features of one embodiment may be beneficially incorporated in other embodiments without further recitation.

DETAILED DESCRIPTION

(8) Embodiments of the present disclosure generally relate to optical devices for augmented, virtual, and mixed reality. More specifically, embodiments described herein provide for methods of forming optical device structures having continuously increasing or decreasing duty cycles. In one embodiment, a device is provided. The device includes a plurality of optical device structures. The plurality of optical device structures have a critical dimension across a length of each optical device structure of the plurality of optical device structures. Each optical device structure of the plurality of optical device structures includes a plurality of discrete zones. The device further includes a plurality of transition zones disposed between two discrete zones of the plurality of discrete zones. The critical dimension of each transition zone of the plurality of transition zones continuously increases or decreases across the length of each optical device structure of the plurality of optical device structures.

(9) FIG. 1 illustrates a perspective, frontal view of an optical device **100**. In one embodiment, which can be combined with other embodiments described herein, the optical device **100** is a waveguide combiner, such as an augmented reality waveguide combiner. In another embodiment, which can be combined with other embodiments described herein, the optical device **100** is a flat optical device, such as a metasurface. The optical device **100** includes a plurality of optical device structures **102** disposed in or on a substrate **101**. The optical device structures **102** may be nanostructures having sub-micron dimensions, e.g., nano-sized dimensions, such as critical dimensions less than 1 μm . In one embodiment, which can be combined with other embodiments described herein, regions of the plurality of optical device structures **102** correspond to one or more gratings **104**, such as a first grating **104a**, a second grating **104b**, and a third grating **104c**. In one embodiment, which can be combined with other embodiments described herein, the optical device **100** is a waveguide combiner that includes at least the first grating **104a** corresponding to an input coupling grating and the third grating **104c** corresponding to an output coupling grating. The waveguide combiner according to the embodiment, which can be combined with other embodiments described herein, may include the second grating **104b** corresponding to an intermediate grating. A portion **105** of the optical device **100** is further described in FIG. 2B.

(10) In one embodiment, which can be combined with other embodiments described herein, the optical device **100** is a master for an imprint process, such as a nano-imprint lithography process. In another embodiment, which can be combined with other embodiments described herein, the optical device **100** is patterned directly, such as via a direct etch process. As described herein, the optical device **100** is one of a master pattern of a master for an imprint process for optical device fabrication (e.g., nano-imprint lithography using a master to form an optical device) or an optical device (e.g., a waveguide combiner and flat optical device). As described herein, the optical device **100** is formed from at least one of a device material or substrate.

(11) In some embodiments, which can be combined with other embodiments described herein, the plurality of optical device structures **102** can be formed from the substrate **101**. The substrate **101** may be formed from any suitable material, provided that the substrate **101** can adequately transmit light in a desired wavelength or wavelength range and can serve as an adequate support for the optical device **100**, described herein. In some embodiments, which can be combined with other embodiments described herein, the material of the substrate **101** has a refractive index that is relatively low, as compared to the refractive index of the plurality of optical device structures **102**. Substrate selection may include substrates of any suitable material, including, but not limited to,

amorphous dielectrics, non-amorphous dielectrics, crystalline dielectrics, silicon oxide, polymers, and combinations thereof. In some embodiments, which may be combined with other embodiments described herein, the substrate **101** includes a transparent material. In one example, the substrate **101** includes silicon (Si), silicon dioxide (SiO₂), germanium (Ge), silicon germanium (SiGe), InP, GaAs, GaN, fused silica, quartz, sapphire, and high-index transparent materials such as high-refractive-index glass.

(12) In some embodiments, which can be combined with other embodiments described herein, the plurality of optical device structures **102** can be formed from a device material. The device material is disposed over the substrate **101**. The device material includes, but is not limited to, one or more of silicon carbide (SiC), silicon oxycarbide (SiOC), titanium dioxide (TiO₂), silicon dioxide (SiO₂), vanadium (IV) oxide (VO_x), aluminum oxide (Al₂O₃), aluminum-doped zinc oxide (AZO), indium tin oxide (ITO), tin dioxide (SnO₂), zinc oxide (ZnO), tantalum pentoxide (Ta₂O₅), silicon nitride (Si₃N₄), zirconium dioxide (ZrO₂), niobium oxide (Nb₂O₅), cadmium stannate (Cd₂SnO₄), silicon carbon-nitride (SiCN) containing materials, combinations thereof, or other suitable materials.

(13) FIG. 2A is a schematic, top-view of a plurality of optical device structures **202** according to embodiments described herein. The plurality of optical device structures **202** include a plurality of discrete zones **204**. As shown in FIG. 2A, the plurality of optical device structures **202** have a first discrete zone **204a**, a second discrete zone **204b**, and a third discrete zone **204c**. Although only three discrete zones **204** of the plurality of discrete zones **204** are shown, more than three discrete zones **204** may be included in the plurality of optical device structures **202**. The plurality of optical device structures **202** may be parallel, normal, or angled with respect to the boundaries between adjacent discrete zones **204** of the plurality of discrete zones **204**.

(14) The plurality of discrete zones **204** have a critical dimension **206**. The critical dimension **206** is the width of each of the plurality of optical device structures **202**. In one embodiment, which can be combined with other embodiments described herein, the first discrete zone **204a** has a first critical dimension **206a**, the second discrete zone **204b** has a second critical dimension **206b**, and the third discrete zone **204c** has a third critical dimension **206c**. In one embodiment, which can be combined with other embodiments described herein, the critical dimension **206** is between about 40 nm and about 380 nm.

(15) Adjacent optical device structures **202** of the plurality of optical device structures **202** have a pitch **208**. The pitch **208** is the distance between leading edges of the adjacent optical device structures **202**. The pitch **208** between adjacent optical device structures **202** remains constant across a length L of the plurality of optical device structures **202**. In one embodiment, which can be combined with other embodiments described herein, the pitch **208** is between about 280 nm and about 450 nm.

(16) Adjacent optical device structures **202** of the plurality of optical device structures **202** have a duty cycle. The duty cycle is determined by dividing the critical dimension **206** of the optical device structure **202** by the pitch **208**. Therefore, the plurality of discrete zones **204** have different duty cycles. The duty cycle changes across the length L of the plurality of optical device structures **202** as the critical dimension **206** is changed. In one embodiment, which can be combined with other embodiments described herein, the duty cycle of the plurality of discrete zones **204** is between about 0.2 to about 0.85. The difference in the duty cycle between adjacent discrete zones **204** of the plurality of discrete zones **204** is greater than 0.01. The critical dimension **206** of each discrete zone **204** of the plurality of discrete zones **204** is constant across the entire zone. The difference in the critical dimensions **206** and the duty cycle between the plurality of discrete zones **204** results in decreased optical device performance.

(17) FIG. 2B is a schematic, top-view of a plurality of optical device structures **102** of a portion **105** of an optical device **100**. In one embodiment, which can be combined with other embodiments described herein, the plurality of optical device structures **102** are of a flat optical device, such as a

metasurface. In another embodiment, which can be combined with other embodiments described herein, the plurality of optical device structures **102** are of a waveguide combiner, such as an augmented reality waveguide combiner. The waveguide combiner according to the embodiment, which can be combined with other embodiments described herein, may include the plurality of optical device structures **102** in at least one of the gratings **104**. In yet another embodiment, which can be combined with other embodiments described herein, the plurality of optical device structures **102** correspond to the first grating **104a**, the second grating **104b**, or the third grating **104c** of the optical device **100**. Although only four of the plurality of optical device structures **102** are shown, one or more of the plurality of optical device structures **102** may be disposed on the optical device **100**.

(18) The plurality of optical device structures **102** have a critical dimension **206**. The critical dimension **206** is the width of each of the plurality of optical device structures **202**. The plurality of optical device structures **102** include a plurality of discrete zones **204**. As shown in FIG. 2B, the plurality of optical device structures **102** have a first discrete zone **204a**, a second discrete zone **204b**, and a third discrete zone **204c**. Although three discrete zones **204** are shown in FIG. 2B, more than three discrete zones **204** may be included on the plurality of optical device structures **102**. The first discrete zone **204a** has a first critical dimension **206a**, the second discrete zone **204b** has a second critical dimension **206b**, and the third discrete zone **204c** has a third critical dimension **206c**. In one embodiment, which can be combined with other embodiments described herein, each discrete zone **204** of the plurality of discrete zones **204** has a critical dimension **206** that continuously increases or decreases across a length L of the plurality of optical device structures **102**. In one example of this embodiment, as shown in FIG. 2B, the first critical dimension **206a**, the second critical dimension **206b**, and the third discrete zone **204c** increase across the length L of the plurality of optical device structures **102**. In some embodiments, as shown in FIG. 2B, the plurality of optical device structures **102** may have a leading edge and a trailing edge opposite the leading edge. In some embodiments, each leading edge and each trailing edge curve outward in an opposite direction from each other along a length L of the plurality of optical device structures **102**.

(19) The plurality of optical device structures **102** further include a plurality of transition zones **210**. The plurality of transition zones **210** are disposed between two discrete zones **204** of the plurality of discrete zones **204**. In one embodiment, which can be combined with other embodiments described herein, a first transition zone **210a** is disposed between the first discrete zone **204a** and the second discrete zone **204b**. A second transition zone **210b** is disposed between the second discrete zone **204b** and the third discrete zone **204c**. Although two transition zones **210** are shown in FIG. 2B, more than two transition zones **210** may be included on the plurality of optical device structures **102**.

(20) In one embodiment, which can be combined with other embodiments described herein, the critical dimension **206** in the plurality of transition zones **210** continuously increases or decreases across the length L of the plurality of optical device structures **102**. For example, the critical dimension **206** continuously increases from the first critical dimension **206a** to the second critical dimension **206b** in the first transition zone **210a**. Additionally, the critical dimension **206** continuously increases from the second critical dimension **206b** to the third critical dimension **206c** in the second transition zone **210b**.

(21) Adjacent optical device structures **102** of the plurality of optical device structures **102** have a pitch **208**. The pitch **208** is the distance between leading edges of the adjacent optical device structures **102**. The pitch **208** between adjacent optical device structures **102** remains constant across the length L of the plurality of optical device structures **102**. In one embodiment, which can be combined with other embodiments described herein, the pitch **208** is between about 280 nm and about 450 nm.

(22) Adjacent optical device structures **102** of the plurality of optical device structures **102** have a duty cycle. The duty cycle is determined by dividing the critical dimension **206** of the optical

device structure **102** by the pitch **208**. In other words, the duty cycle is defined as the critical dimension **206** of the plurality of optical device structures **102** divided by the pitch of adjacent optical device structures **102** of the plurality of optical device structures **102**. The duty cycle increases or decreases across the length **L** of the plurality of optical device structures **102** as the critical dimension **206** is changed. Therefore, as shown in FIG. 2B, the duty cycle will continuously increase or decrease across the length **L**. The critical dimension **206** of the plurality of optical device structures **102** continuously increasing or decreasing across the length **L** improves the optical performance of the optical device **100**.

(23) FIGS. 3A and 3B schematic, cross-sectional views of a substrate **101**. FIG. 4 is a flow diagram of a method **400** for forming a plurality of optical device structures **102**. At operation **401**, as shown in FIG. 3A, a photoresist **304** is deposited. The photoresist **304** is deposited over a hardmask **302**. The photoresist **304** is sensitive to electromagnetic radiation. The photoresist **304** can be a positive or negative photoresist.

(24) A positive photoresist includes portions of the photoresist **304** that, when exposed to radiation, are respectively soluble to a photoresist developer applied to the photoresist **304** after the pattern is written into the photoresist **304** using the electromagnetic radiation. A negative photoresist includes portions of the photoresist **304** that, when exposed to radiation, will be respectively insoluble to the photoresist developer applied to the photoresist **304** after the pattern is written into the photoresist **304** using the electromagnetic radiation. The chemical composition of the photoresist **304** determines whether the photoresist is a positive photoresist or negative photoresist. Examples of the photoresist **304** include, but are not limited to, at least one of diazonaphthoquinone, a phenol formaldehyde resin, poly(methyl methacrylate), poly(methyl glutarimide), and SU-8. The hardmask **302** includes, but is not limited to, silicon, silicon nitride (SiN), SiO, low-k, SiC, SiOC, SiCONH, TaO, BPSG, PSG, dielectrics, spin on glass (SOG), or metallic alloys such as TiN.

(25) At operation **402**, as shown in FIG. 3B, the photoresist **304** is patterned. Prior to patterning the photoresist **304**, a design file is edited. FIG. 3C is a schematic, top-view of a mask **308**. As shown in FIG. 3C, the design file corresponds to the mask **308** to be developed. The design file is edited such that the mask **308** will include a width **312** of a plurality of openings **310** that continuously increases or decreases across a length **L**. In one embodiment, which can be combined with other embodiments described herein, an optical proximity correction (OPC) step is performed on the design file. The OPC step is utilized to meet a critical dimension **206** of a plurality of discrete zones **204** and the critical dimension **206** of a plurality of transition zones **210**.

(26) The mask **308** is generated. The mask **308** is generated with the plurality of openings **310** corresponding to the plurality of optical device structures **102** to be patterned. Light passes through the plurality of openings **310** in the mask **308** to expose the photoresist **304** to electromagnetic radiation. The plurality of openings **310** have identical or substantially identical shapes as the plurality of optical device structures **102** to be patterned. The width **312** corresponds to the critical dimension **206** of the plurality of optical device structures **102**. In one embodiment, which can be combined with other embodiments described herein, the width **312** of each opening **310** of the plurality of openings **310** is larger than the critical dimension **206** at the corresponding point along the length **L**. The width **312** of each opening **310** of the plurality of openings **310** is larger than the critical dimension **206** at the corresponding point along the length **L** to account for a decrease of the critical dimension **206** compared to the width **312** during the method **400**. For example, a patterned photoresist **306** to be formed after light is passed through the plurality of openings **310** will have smaller dimensions than the width **312** of each opening **310**. In another embodiment, which can be combined with other embodiments described herein, the width **312** of each opening **310** of the plurality of openings **310** is smaller than the critical dimension **206** at the corresponding point along the length **L**.

(27) As shown in FIG. 3B, after exposure of the photoresist **304** to the electromagnetic radiation, the photoresist **304** is developed to leave a patterned photoresist **306** on the hardmask **302**. At

operation **403**, the hardmask **302** is etched. Using the patterned photoresist **306**, the hardmask **302** is pattern etched through the openings in the patterned photoresist **306**. The hardmask **302** includes, but is not limited to, silicon nitride (SiN), SiO, low-k, SiC, SiOC, SiCONH, TaO, BPSG, PSG, dielectrics, or metallic alloys such as TiN.

(28) At operation **404**, the plurality of optical device structures **102** are formed. The plurality of optical device structures **102** are formed by etching the substrate **101**. In one embodiment, which can be combined with other embodiments described herein, the plurality of optical device structures **102** are formed by etching a device material disposed on the substrate **101**. The substrate **101** or the device material is pattern etched through the openings in the hardmask **302**. As shown in FIG. 2B, the plurality of optical device structures **102** includes a plurality of discrete zones **204** and a plurality of transition zones **210**. In one embodiment, which can be combined with other embodiments described herein, a critical dimension **206** continuously increases or decreases across a length L of the plurality of optical device structures. Additionally, the duty cycle between the plurality of discrete zones **204** and the plurality of transition zones **210** continuously increases or decreases across the length L.

(29) In summation, methods of forming optical device structures having continuously increasing or decreasing duty cycles are disclosed. The optical device structures each have a plurality of transition zones disposed between two discrete zones. The critical dimensions of each transition zone of the plurality of transition zones continuously increase or decrease across a length of each optical device structure. Adjacent optical device structures include a constant pitch. A duty cycle defined as the critical dimension of the plurality of optical device structures divided by the pitch continuously increases or decreases across the length of each optical device structure. The continuous increasing or decreasing across the length of each optical device structure improves the optical performance of the optical device structures. The performance is improved as the continuous transition between two discrete zones allows for the light to be propagated more efficiently in the optical devices.

(30) While the foregoing is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A device, comprising: a plurality of optical device structures, each optical device structure having a leading edge and a trailing edge opposite the leading edge, wherein each leading edge and each trailing edge curve outward in an opposite direction from each other along a length of each optical device structure, the plurality of optical device structures having a critical dimension across the length of each optical device structure of the plurality of optical device structures, each optical device structure of the plurality of optical device structures having: a plurality of discrete zones; a plurality of transition zones disposed between two discrete zones of the plurality of discrete zones, the critical dimension of each transition zone of the plurality of transition zones continuously increasing or decreasing across the length of each optical device structure of the plurality of optical device structures; and a duty cycle defined as the critical dimension of the plurality of optical device structures divided by a pitch of adjacent optical device structures of the plurality of optical device structures, the duty cycle continuously increasing or decreasing across the length of each optical device structure of the plurality of optical device structures, wherein the pitch is a distance between leading edges of the adjacent optical device structures and the pitch remains constant across the length of each optical device structure of the adjacent optical device structures, and wherein the critical dimension is a width of each optical device structure of the plurality of optical device structures.

2. The device of claim 1, wherein the plurality of optical device structures correspond to one or

more gratings or a master pattern of a master.

3. The device of claim 2, wherein the one or more gratings include an input coupling grating, an output coupling grating, and an intermediate grating.

4. The device of claim 1, wherein the pitch is between about 280 nm and about 450 nm.

5. The device of claim 1, wherein the plurality of optical device structures are formed on a substrate, the substrate includes one or more of silicon (Si), silicon dioxide (SiO₂), germanium (Ge), silicon germanium (SiGe), InP, GaAs, GaN, fused silica, quartz, sapphire, or high-index transparent materials.

6. The device of claim 1, wherein the plurality of optical device structures are formed from a device material, the device material includes one or more of silicon carbide (SiC), silicon oxycarbide (SiOC), titanium dioxide (TiO₂), silicon dioxide (SiO₂), vanadium (IV) oxide (VOx), aluminum oxide (Al₂O₃), aluminum-doped zinc oxide (AZO), indium tin oxide (ITO), tin dioxide (SnO₂), zinc oxide (ZnO), tantalum pentoxide (Ta₂O₅), silicon nitride (Si₃N₄), zirconium dioxide (ZrO₂), niobium oxide (Nb₂O₅), cadmium stannate (Cd₂SnO₄), or silicon carbon-nitride (SiCN) containing materials.

7. The device of claim 1, wherein a first critical dimension, a second critical dimension, and a third critical dimension of the plurality of discrete zones increase or decrease across the length of the plurality of optical device structures.

8. A device, comprising: a plurality of optical device structures, each optical device structure having a leading edge and a trailing edge opposite the leading edge, wherein each leading edge and each trailing edge curve outward in an opposite direction from each other along a length of each optical device structure, the plurality of optical device structures having a critical dimension across the length of each optical device structure of the plurality of optical device structures, each optical device structure of the plurality of optical device structures having: a plurality of discrete zones; a plurality of transition zones disposed between two discrete zones of the plurality of discrete zones, the critical dimension of each transition zone of the plurality of transition zones continuously increasing or decreasing across the length of each optical device structure of the plurality of optical device structures, adjacent optical device structures of the plurality of optical device structures including a pitch; and a duty cycle defined as the critical dimension of the plurality of optical device structures divided by the pitch, the duty cycle continuously increasing or decreasing across the length of each optical device structure of the plurality of optical device structures, wherein the critical dimension is defined by a width of each optical device structure of the plurality of optical device structures, wherein the pitch is a distance between leading edges of the adjacent optical device structures and the pitch remains constant across the length of each optical device structure of the adjacent optical device structures, and wherein the critical dimension of at least one discrete zone of the plurality of discrete zones is constant across the length of the discrete zone.

9. The device of claim 8, wherein the plurality of optical device structures correspond to one or more gratings or a master pattern of a master.

10. The device of claim 8, wherein the pitch is between about 280 nm and about 450 nm.

11. A method, comprising: editing a design file, the design file corresponding to a mask, the mask including a plurality of openings, the plurality of openings having a width that continuously increases or decreases across a length of the plurality of openings; generating the mask, the mask to be disposed over a photoresist disposed over a substrate; passing light through the plurality of openings of the mask positioned over the photoresist; developing the photoresist to form a patterned photoresist, the patterned photoresist corresponding to the plurality of openings of the mask; and patterning the substrate or a device material to form a plurality of optical device structures, each optical device structure having a leading edge and a trailing edge opposite the leading edge, wherein each leading edge and each trailing edge curve outward in an opposite direction from each other along a length of each optical device structure, the optical device structures corresponding to the plurality of openings of the mask, the plurality of optical device

structures having a critical dimension that continuously increases or decreases across the length of each optical device structure of the plurality of optical device structures, each optical device structure of the plurality of optical device structures having: a plurality of discrete zones; a plurality of transition zones disposed between two discrete zones of the plurality of discrete zones, the critical dimension of each transition zone of the plurality of transition zones continuously increasing or decreasing across the length of each optical device structure of the plurality of optical device structures; and a duty cycle defined as the critical dimension of the plurality of optical device structures divided by a pitch of adjacent optical device structures of the plurality of optical device structures, the duty cycle continuously increasing or decreasing across the length of each optical device structure of the plurality of optical device structures, wherein the pitch is a distance between leading edges of the adjacent optical device structures and the pitch remains constant across the length of each optical device structure of the adjacent optical device structures, and wherein the critical dimension is a width of each optical device structure of the plurality of optical device structures.

12. The method of claim 11, wherein the photoresist is disposed on a hardmask, the hardmask disposed on a substrate.

13. The method of claim 12, wherein the device material is disposed between the substrate and the hardmask.

14. The method of claim 13, further comprising pattern etching the hardmask through openings in the patterned photoresist.

15. The method of claim 14, wherein the device material includes one or more of silicon carbide (SiC), silicon oxycarbide (SiOC), titanium dioxide (TiO₂), silicon dioxide (SiO₂), vanadium (IV) oxide (VOx), aluminum oxide (Al₂O₃), aluminum-doped zinc oxide (AZO), indium tin oxide (ITO), tin dioxide (SnO₂), zinc oxide (ZnO), tantalum pentoxide (Ta₂O₅), silicon nitride (Si₃N₄), zirconium dioxide (ZrO₂), niobium oxide (Nb₂O₅), cadmium stannate (Cd₂SnO₄), or silicon carbon-nitride (SiCN) containing materials.

16. The method of claim 11, wherein the substrate includes one or more of silicon (Si), silicon carbide (SiC), silicon dioxide (SiO₂), germanium (Ge), silicon germanium (SiGe), InP, GaAs, GaN, fused silica, quartz, sapphire, or high-index transparent materials.

17. The method of claim 11, wherein the plurality of optical device structures correspond to one or more gratings or a master pattern of a master.

18. The method of claim 17, wherein the one or more gratings include an input coupling grating, an output coupling grating, and an intermediate grating.

19. The method of claim 11, wherein the pitch is between about 280 nm and about 450 nm.

20. The method of claim 11, wherein the width of each opening of the plurality of openings is larger than the critical dimension at a corresponding point of the length of each optical device structure.
